# Necessary and Sufficient Conditions for the Existence of an LU Factorization

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**Abstract**

We establish necessary and sufficient conditions for the existence of an LU factorization $A = LU$ for an arbitrary square matrix $A$, including singular and rank-deficient cases, without the use of row or column permutations. We prove that such a factorization exists if and only if the nullity of every leading principal submatrix is bounded by the sum of the nullities of the corresponding leading column and row blocks. While building upon the work of Okunev and Johnson, we present simpler, constructive proofs based on induction. Furthermore, we extend these results to characterize rank-revealing factorizations, providing explicit sparsity bounds for the factors $L$ and $U$. Finally, we derive analogous necessary and sufficient conditions for the existence of factorizations constrained to have unit lower or unit upper triangular factors.

It is well-known that for any square matrix $A$, there exists a permutation of rows (and potentially columns) such that an LU factorization exists. Specifically, the factorization $PA = LU$, where $P$ is a permutation matrix, $L$ is unit lower triangular, and $U$ is upper triangular, is guaranteed to exist for any matrix $A$ [2, 5].

However, the existence of an LU factorization for a matrix $A$ *without* permutations—that is, $A = LU$—is not guaranteed. A classical result states that for a non-singular matrix, such a factorization exists if and only if all its leading principal minors are non-zero. When the matrix is singular or when leading principal minors vanish, the situation becomes significantly more complex. The standard conditions involving determinants are no longer sufficient to characterize the existence of the factorization.

This paper addresses the necessary and sufficient conditions for the existence of an LU factorization for an arbitrary matrix $A$, without imposing restrictions on its rank or invertibility. We extend the seminal work of Okunev and Johnson [7], who established rank-based conditions for this problem (see also [1] for a discussion on multiple factorizations of singular matrices). Other perspectives on factorizations and their existence in general algebraic structures or specific contexts can be found in [6, 4, 3].

The contributions of this paper are threefold. First, we formulate the necessary and sufficient condition using the *nullity* of the leading principal submatrices relative to the nullity of the corresponding column and row blocks. Specifically, we prove that an LU factorization exists if and only if for all $k$:

$$\mathrm{null}(A[1:k, 1:k]) \leq \mathrm{null}(A[:, 1:k]) + \mathrm{null}(A[1:k, :]^T)$$

Second, we provide new, simpler proofs of these results. Unlike previous algebraic derivations, our proofs are constructive and rely on induction, explicitly building the factorization by handling the singular cases via internal permutations that preserve the triangular structure. Finally, we extend these results to characterize rank-revealing factorizations, providing tight sparsity bounds for the factors, and we derive analogous conditions for factorizations requiring unit lower or unit upper triangular factors.

The remainder of the paper is organized as follows. Section 1 reviews classical results regarding factorizations with permutations. Section 2 presents the main theorem and the derivation of the nullity condition. Theorem 5 strengthens this result by characterizing rank-revealing factorizations. Finally, Section 3 discusses the specific requirements for unit triangular factors.

We begin with a few classical results that will serve as reference for the later sections.

## Notations

| | |
|---|---|
| $\mathrm{rank}(A)$ | Rank of matrix $A$ |
| $\mathrm{null}(A)$ | Dimension of the null space of matrix $A$ |
| $A^T$ | Transpose of matrix $A$ |
| $a_{ij}$ | Entry $(i, j)$ of matrix $A$ |
| $A[:, i]$ | Column $i$ of matrix $A$ |
| $A[i, :]$ | Row $i$ of matrix $A$ |
| $A[i : j, k : l]$ | Submatrix of $A$ with rows from $i$ to $j$ and columns from $k$ to $l$ |
| $A[i : j, k :]$ | Submatrix of $A$ with rows from $i$ to $j$ and columns from $k$ to the end |
| $e_i$ | $i$-th vector of the standard basis |
| $I_n$ | Identity matrix of size $n$ |

# 1 Classical results

The theorems establish that any matrix can be permuted to have an LU factorization with unit triangular factors.

**Theorem 1.** *Consider matrix $A$ of size $n$. There exist a row permutation matrix $P$ such that $PA = LU$ where $L$ is unit lower triangular and $U$ is upper triangular.*

*Proof.* This is a classical result. We can prove it by induction on $n$. The result is true for $n = 1$.

Assume it is true for $n - 1$. If $a_{11} \neq 0$, then consider:

$$A = LDU$$

where

$$L = \begin{pmatrix} 1 & 0 \\ a_{11}^{-1} A_{21} & I \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} a_{11} & A_{12} \\ 0 & I \end{pmatrix}$$

We can apply the induction hypothesis to $S$. The result for $A$ follows.

If $a_{11} = 0$, we have two cases.

Case 1: column 1 of $A$ is non-zero. We can find the smallest $i$ such that $A[i, 1] \neq 0$. Pivot row 1 and $i$. Denote by $B = \Pi_i A$ the permuted matrix (with $\Pi_i^2 = I$). Since $b_{11} \neq 0$, $B$ has an LU factorization. This proves the result for $A$.

Case 2: column 1 of $A$ is zero. We choose:

$$A = IDU$$

where

$$D = \begin{pmatrix} 1 & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} 0 & A_{12} \\ 0 & I \end{pmatrix}$$

We can apply the induction hypothesis to $S$. The result for $A$ follows. □

**Corollary 1.** *Consider matrix $A$ of size $n$. There exist a column permutation matrix $Q$ such that $AQ = LU$ where $L$ is lower triangular and $U$ is unit upper triangular.*

*Proof.* Apply Theorem 1 to $A^T$. □

**Definition 1.** *Matrix $A$ of size $m \times n$ is said to have a rank revealing LU factorization if $A = LU$ where $L$ is lower triangular of size $m \times r$, $U$ is upper triangular of size $r \times n$, and $r = \mathrm{rank}(A)$.*

If we use both row and column permutations, we can always obtain a rank revealing LU factorization.

**Theorem 2.** *Consider matrix $A$ of size $n$. There exist a row permutation matrix $P$ and a column permutation matrix $Q$ such that $PAQ = LU$ is a rank revealing $LU$ factorization.*

*Proof.* This is also a classical result. We can prove it by induction on $n$. The result is true for $n = 1$.

Assume it is true for $n-1$. If $a_{11} \neq 0$, then consider:

$$A = LDU$$

where

$$L = \begin{pmatrix} a_{11} & 0 \\ A_{21} & I \end{pmatrix}, \quad D = \begin{pmatrix} a_{11}^{-1} & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} a_{11} & A_{12} \\ 0 & I \end{pmatrix}$$

$S$ has rank $r-1$ if $r = \text{rank}(A)$. We can apply the induction hypothesis to $S$. The result for $A$ then follows.

If $a_{11} = 0$, we have two cases. If $A$ is 0, we pick $L = U = 0$. Otherwise we have some $i$ and $j$ such that $a_{ij} \neq 0$. We can permute row 1 and $i$ and column 1 and $j$. Denote by $B = \Pi_i A \Pi_j$ the permuted matrix. Since $b_{11} \neq 0$, $B$ has a rank revealing LU factorization. This proves the result for $A$.

We have also proved that this process must terminate after $r$ steps where $r = \text{rank}(A)$. So the LU factorization of $B$ is rank revealing.

From Lemma 3 and Corollary 4, we can also prove that the remaining Schur complement is zero after $r$ steps of the LU factorization. □

We now consider LU factorizations without permutations. The simplest result requires that all leading principal minors are non-singular.

**Proposition 1.** *Consider an invertible matrix $A$ of size $n$. Matrix $A$ has an LU factorization if and only if*

$$\text{rank}(A[1:k, 1:k]) = k, \quad k = 1, \ldots, n$$

*Proof.* We prove that the condition is necessary. Assume $A = LU$:

$$\det(A) = \prod_{i=1}^{n} l_{ii} u_{ii}$$

and $\det(A) \neq 0$. So, $l_{ii} \neq 0$ and $u_{ii} \neq 0$ for all $i$. Therefore: $\text{rank}(A[1:k, 1:k]) = k$, $k = 1, \ldots, n$.

We prove that the condition is sufficient. We prove the result by induction on $n$. The result is true for $n = 1$.

Assume it is true for $n-1$. By assumption, $a_{11} \neq 0$. We then have the factorization:

$$A = LDU$$

where

$$L = \begin{pmatrix} a_{11} & 0 \\ A_{21} & I \end{pmatrix}, \quad D = \begin{pmatrix} a_{11}^{-1} & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} a_{11} & A_{12} \\ 0 & I \end{pmatrix}$$

and $S = A_{22} - A_{21} a_{11}^{-1} A_{12}$ is the Schur complement. $L$ and $U$ are full rank. So $D$ satisfies the condition of the theorem. Consequently, $S$ also satisfies the condition of the theorem and is of size $n-1$. By the induction hypothesis, $S$ has an LU factorization. Therefore $A$ also has an LU factorization. □

## 2 General necessary and sufficient condition for the existence of an LU factorization

We now discuss the core of this paper: the necessary and sufficient conditions for the existence of an LU factorization without permutations. We start with an illustrative example that will preview the main result.

Consider matrix

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Matrix $A$ does not have an LU factorization. We prove this by contradiction. Assume $A = LU$. Then $l_{11} u_{11} = 0$. Assume $l_{11} = 0$. Then $l_{11} u_{12} = 0 = 1$. Contradiction. Assume $u_{11} = 0$. Then $l_{21} u_{11} = 0 = 1$. Contradiction.

Consider now

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

This matrix in fact has an LU factorization. We can find it using the following procedure, which we will detail later on for the general case. Surprisingly the process involves row and column permutations. However the final result remains an LU factorization without the permutations.

Now, perform a cyclic shift of the rows: (1,2,3) $\to$ (2,3,1) and permute column 1 and 3:

$$PAQ = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

But surprisingly we have:

$$A = P^{-1} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} Q^{-1} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

This is valid rank revealing LU factorization of $A$. Similarly:

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

is another valid rank revealing LU factorization of $A$.

We can generalize this example. Consider any $B = LU$ and:

$$A = \begin{pmatrix} 0_n & 0_n & 0_n \\ 0_n & 0_n & C \\ 0_n & B & D \end{pmatrix} = \begin{pmatrix} 0_n & 0_n \\ C & 0_n \\ D & L \end{pmatrix} \begin{pmatrix} 0_n & 0_n & I_n \\ 0_n & U & 0_n \end{pmatrix}$$

If $C = LU$ then:

$$A = \begin{pmatrix} 0_n & 0_n \\ 0_n & L \\ I_n & 0_n \end{pmatrix} \begin{pmatrix} 0_n & B & D \\ 0_n & 0_n & U \end{pmatrix}$$

Of course, this is not true for all matrices. We will show that Property 1 below is a necessary and sufficient condition for the existence of an LU factorization. The proof will rely on constructing a $P$ and $Q$ such that $B = P^{-1}AQ^{-1}$. $B[1:r, 1:r]$ will satisfy the conditions of Proposition 1 where $r = \operatorname{rank}(A) = \operatorname{rank}(B)$. Therefore $B = LU$. From this, we will deduce a rank-revealing LU factorization of $A = (PL)(UQ)$, where $PL$ is lower triangular with $r$ columns, and $UQ$ is upper triangular with $r$ rows.

**Property 1.** *Matrix $A$ is square of size $n$. For all $k = 1, \ldots, n$:*

$$\operatorname{null}(A[1:k, 1:k]) \le \operatorname{null}(A[:, 1:k]) + \operatorname{null}(A[1:k, :]^T) \tag{1}$$

**Theorem 3.** *The rank-nullity theorem states that for any matrix $A$ of size $m \times n$:*

$$\operatorname{rank}(A) + \operatorname{null}(A) = n$$

We now state Sylvester's inequality for the rank of a product of matrices.

**Lemma 1.** *Consider $A = BC$. Then:*

$$\operatorname{rank}(A) \le \min(\operatorname{rank}(B), \operatorname{rank}(C)), \quad \operatorname{rank}(B) + \operatorname{rank}(C) - k \le \operatorname{rank}(A)$$

*where $k$ is the number of columns of $B$ and rows of $C$.*

**Corollary 2.** *If $A$, $B$, and $C$ are square matrices then:*

$$\text{null}(A) \leq \text{null}(B) + \text{null}(C)$$

*Proof.* Use Lemma 1 and Theorem 3. $\square$

Assume that $A$, a square matrix of size $n$, has an LU factorization:

$$A = LU$$

Then:

$$\begin{aligned} A[1:k,:] &= L[1:k,1:k]U[1:k,:], \\ A[:,1:k] &= L[:,1:k]U[1:k,1:k], \\ A[1:k,1:k] &= L[1:k,1:k]U[1:k,1:k] \end{aligned}$$

Using Lemma 1 and Corollary 2, we have:

$$\begin{aligned} \text{rank}(A[1:k,:]) &\leq \text{rank}(L[1:k,1:k]), \\ \text{rank}(A[:,1:k]) &\leq \text{rank}(U[1:k,1:k]), \\ \text{null}(A[1:k,1:k]) &\leq \text{null}(L[1:k,1:k]) + \text{null}(U[1:k,1:k]) \end{aligned}$$

Using Theorem 3, we get:

$$\text{null}(A[1:k,1:k]) \leq \text{null}(A[:,1:k]) + \text{null}(A[1:k,:]^T)$$

So Property 1 is a necessary condition for the existence of an LU factorization.

We will prove the following theorem:

**Theorem 4.** *Matrix $A$ satisfies Property 1 if and only if it has an $LU$ factorization.*

*Proof.* We prove the result by induction on the size of $A$. The result is true for $n = 1$.

Assume it is true for size $n - 1$.

Assume $a_{11} \neq 0$. We have the factorization:

$$A = LDU$$

where

$$L = \begin{pmatrix} a_{11} & 0 \\ A_{21} & I \end{pmatrix}, \quad D = \begin{pmatrix} a_{11}^{-1} & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} a_{11} & A_{12} \\ 0 & I \end{pmatrix}$$

Since $L$ and $U$ are full rank, $D$ satisfies Property 1. Consequently, $S$ also satisfies Property 1 and is of size $n-1$. Moreover $S$ has rank $r - 1$ if $r = \text{rank}(A)$. By the induction hypothesis, $S$ has an LU factorization. The result for $A$ follows.

Assume $a_{11} = 0$. From Property 1, we have either:

1. $A[:,1] = 0$, or
2. $A[1,:] = 0$
3. both.

Case 1: $A[:,1] = 0$. We can find the smallest $j$ such that $A[1,j] \neq 0$. Pivot column 1 and $j$. Denote by $B = A\Pi_j$ the permuted matrix (with $\Pi_j^2 = I$). We prove that $B$ satisfies Property 1.

Consider $k < j$. We have

$$\text{null}(A[1:k,1:k]) \leq \text{null}(A[:,1:k]) + \text{null}(A[1:k,:]^T) \tag{2}$$

Since column 1 of $A$ is zero, we have:

$$\text{null}(A[1:k,1:k]) = \text{null}(A[1:k,2:k]) + 1$$

We have

$$\text{null}(B[1:k,2:k]) = \text{null}(A[1:k,2:k])$$

Since the only non-zero entry in row 1 of $B[1:k,1:k]$ is in column 1 we have

$$\text{null}(B[1:k,1:k]) = \text{null}(B[1:k,2:k]) = \text{null}(A[1:k,2:k]) = \text{null}(A[1:k,1:k]) - 1$$

By a similar reasoning, we have

$$\text{null}(B[:,1:k]) = \text{null}(A[:,1:k]) - 1$$

The column pivoting does not affect the rank of the rows, so

$$\text{null}(B[1:k,:]^T) = \text{null}(A[1:k,:]^T)$$

Therefore $B$ satisfies Property 1 for $k < j$.

For $k \geq j$, the pivoting does not affect the ranks in Equation (2), so $B$ also satisfies Property 1 for $k \geq j$.

So $B$ satisfies Property 1.

Since $b_{11} \neq 0$, $B$ has an LU factorization: $B = LU$. Column $j$ of $B$ is zero. If column $j$ of $U$ is non-zero, we can zero it out without affecting any column in $LU$. So we can assume that column $j$ of $U$ is zero. We have:

$$A = B\Pi_j = LU\Pi_j$$

and $U\Pi_j$ is still upper triangular because $\Pi_j$ only permutes column 1 and $j$. Therefore $A$ has an LU factorization.

Case 2: $A[1,:] = 0$. Consider $B = A^T$. $B$ corresponds to Case 1, so $B$ has an LU factorization: $B = LU$. Therefore:

$$A = B^T = U^T L^T$$

is an LU factorization of $A$.

Case 3: If $A$ is 0, it has an LU factorization with $L = U = 0$. Let's denote by $j$ the smallest index such that column $j$ is non zero. Let's denote by $i > 1$ the smallest index such that $a_{ij} \neq 0$. Let's pivot column 1 and $j$ and denote by $B = A\Pi_j$ the permuted matrix. $B$ satisfies Case 2, so it has an LU factorization: $B = LU$. Column $j$ of $U$ is zero. Therefore:

$$A = B\Pi_j = LU\Pi_j$$

is an LU factorization of $A$.

We have also proved that this process must terminate after $r$ steps where $r = \text{rank}(A)$. So the LU factorization of $B$ is rank revealing. □

We now prove a stronger theorem. Please compare this result with Theorem 2.

**Theorem 5.** *Matrix $A$ is of size $n$ and rank $r \leq n$. If $A$ satisfies Property 1, then there exist a row permutation $P_r$ and column permutation $P_c$ such that:*

$$B = P_r^{-1} A P_c^{-1}$$

*with $B[1:k,1:k]$, $1 \leq k \leq r$, non-singular. $B[1:r,1:r]$ has an LU factorization:*

$$B[1:r,1:r] = L_r U_r$$

*and:*

$$B = \left(B[:,1:r]U_r^{-1}\right)\left(L_r^{-1}B[1:r,:]\right) = L_B U_B$$

*is its rank revealing LU factorization. Then:*

$$A = (P_r L_B)(U_B P_c) = LU$$

*is a rank revealing factorization of $A$. Moreover, we have the following stronger sparsity results. If $e_{i_0} = P_r e_i$, then for $L$:*

- *If $i \le r$, then $i_0 \ge i$ and $L[i_0, i+1:] = 0$.*
- *If $i > r$, then $i_0 \le i$ and $L[i_0, \min(r, i_0):] = 0$.*

*We have similar results for $U$. If $e_{j_0} = P_c^T e_j$, then:*

- *If $j \le r$, then $j_0 \ge j$ and $U[j+1:, j_0] = 0$.*
- *If $j > r$, then $j_0 \le j$ and $U[\min(r, j_0):, j_0] = 0$.*

*Proof.* We follow a proof similar to the one for Theorem 4.

We do an induction on the size of $A$. The result is true for $n = 1$.

Assume it is true for size $n-1$. If $a_{11} \neq 0$, we have the factorization:

$$A = LDU$$

where

$$L = \begin{pmatrix} a_{11} & 0 \\ A_{21} & I \end{pmatrix}, \quad D = \begin{pmatrix} a_{11}^{-1} & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} a_{11} & A_{12} \\ 0 & I \end{pmatrix}$$

We apply the induction hypothesis to $S$. We can define $P_r$ and $P_c$ in terms of the permutations for $S$:

$$P_r = \begin{pmatrix} 1 & 0 \\ 0 & P_{r,S} \end{pmatrix}, \quad P_c = \begin{pmatrix} 1 & 0 \\ 0 & P_{c,S} \end{pmatrix}$$

The sparsity results follow directly from the induction hypothesis.

Assume $a_{11} = 0$. From Property 1, we have either:

1. $A[:, 1] = 0$, or
2. $A[1, :] = 0$
3. both.

Case 1: $A[:, 1] = 0$. As before, we can find the smallest $j$ such that $A[1, j] \neq 0$. We pivot column 1 and $j$. Denote by $B_0 = A\Pi_j$ the permuted matrix. $B_0$ satisfies Property 1 and $b_{11} \neq 0$. So $B_0$ has an LU factorization with the sparsity pattern of Theorem 5. Since column $j$ of $B_0$ is zero, column $j$ of $U_{B_0}$ is zero. We define

$$P_c = P_{c,B}\Pi_j, \quad U = U_B \Pi_j$$

$\Pi_j$ swaps columns 1 and $j$ of $U_{B_0}$. We have $e_1 = \Pi_j^T e_j$ and column $j$ of $U_{B_0}$ is zero. We also have a $j_0$ such that $e_j = P_{c,B}^T e_{j_0}$ and $e_1 = P_c^T e_{j_0}$, with $j_0 > r$. Since column 1 of $U$ is zero and the other columns of $U$ have the sparsity pattern of Theorem 5, $U$ also satisfies the sparsity pattern of Theorem 5.

Case 2: $A[1, :] = 0$. Consider $B = A^T$. $B$ corresponds to Case 1 and satisfies the sparsity pattern of Theorem 5. So $A$ also satisfies the sparsity pattern of Theorem 5.

Case 3: If $A = 0$, it has an LU factorization. In that case we pick $L = U = 0$ as the factors. If $A \neq 0$, let's denote by $j$ the smallest index such that column $j$ is non zero. Let's denote by $i > 1$ the smallest index such that $a_{ij} \neq 0$. Let's pivot column 1 and $j$ and denote by $B = A\Pi_j$ the permuted matrix. $B$ satisfies Case 2, so it has an LU factorization: $B = LU$ with the sparsity pattern of Theorem 5. Following a reasoning analogous to Case 1, we can show that $A$ also satisfies the sparsity pattern of Theorem 5.

As before, we have also proved that this process must terminate after $r$ steps where $r = \text{rank}(A)$. So the LU factorization of $B$ is rank revealing. $\square$

# 3 Unit triangular factors

We now prove results for unit triangular factors. Please compare with Theorem 1.

**Theorem 6.** *Matrix $A$ is of size $n$ and rank $r \leq n$. $A$ satisfies, for all $k = 1, \dots, n$:*

$$\mathrm{null}(A[1:k, 1:k]) = \mathrm{null}(A[:, 1:k]) \tag{3}$$

*if and only if $A$ has an LU factorization $A = LU$ where $L$ is unit lower triangular.*

*If $A$ satisfies Equation (3), then there exist a column permutation $P_c$ such that:*

$$B = AP_c^{-1}$$

*$B$ has an LU factorization: $B = L_B U_B$ where $L_B$ is unit lower triangular. Then:*

$$A = L_B(U_B P_c) = LU$$

*is an LU factorization of $A$ where $L = L_B$ is unit lower triangular.*

*We also have that row $j$ of $B$ (or row $j$ of $A$ since the rows are not permuted) is a linear combination of the previous rows if and only if column $j$ of $B$ is a linear combination of the previous columns.*

*Assume $e_{j_0} = P_c^T e_j$. If row $j$ of $B$ is a linear combination of the previous rows, then $j_0 \leq j$. We have $L[:, j] = e_j$, $U[j, :] = 0$, and $U[j_0 :, j_0] = 0$. If row $j$ of $B$ is linearly independent of the previous rows then $j_0 \geq j$ and $U[j + 1 :, j_0] = 0$.*

*Proof.* We first prove that if $A = LU$ with $L$ unit lower triangular, then $A$ satisfies Equation (3). $U$ trivially satisfies Equation (3) since it is upper triangular. Since $L[1 : k, 1 : k]$ is full rank, $A$ also satisfies Equation (3).

We follow a proof similar to the one for Theorem 5. We do an induction on the size of $A$. The result is true for $n = 1$.

Assume it is true for size $n - 1$. If $a_{11} \neq 0$, we verify that the result is true by the induction hypothesis.

Assume $a_{11} = 0$. From Equation (3), we have $A[:, 1] = 0$.

Case 1: If row 1 of $A$ is non-zero, we can use the same steps as in the proof as Theorem 5 with a column permutation to prove the result by induction. In addition, this shows that if $e_{j_0} = P_c^T e_j$ and if row $j$ of $B$ is linearly independent of the previous rows then $j_0 \geq j$ and $U[j + 1 :, j_0] = 0$.

Case 2: If row 1 of $A$ is zero, we cannot use the same proof as Theorem 5 because we can no longer permute rows. However, since row 1 of $A$ is zero, we can define:

$$A = IDU$$

with

$$D = \begin{pmatrix} 1 & 0 \\ 0 & S \end{pmatrix}, \quad U = \begin{pmatrix} 0 & 0 \\ 0 & I \end{pmatrix}$$

$D$ satisfies

$$\mathrm{null}(D[1 : k, 1 : k]) = \mathrm{null}(A[1 : k, 1 : k]) - 1 = \mathrm{null}(A[:, 1 : k]) - 1 = \mathrm{null}(D[:, 1 : k])$$

$D$ satisfies Equation (3). By the induction hypothesis, we can find a column permutation $P_{c,S}$ such that $SP_{c,S}^{-1}$ has an LU factorization with unit lower triangular factor.

This also proves that row $j$ of $B$ is a linear combination of the previous rows if and only if column $j$ of $B$ is a linear combination of the previous columns. Moreover, if $e_{j_0} = P_c^T e_j$ and row $j$ of $B$ is a linear combination of the previous rows, then $j_0 \leq j$. We have $L[:, j] = e_j$, $U[j, :] = 0$, and $U[j_0 :, j_0] = 0$.

Note that the step above with $D$ is not a valid step in Theorem 5. Indeed:

$$\begin{aligned} \mathrm{null}(D[1 : k, 1 : k]) &= \mathrm{null}(A[1 : k, 1 : k]) - 1 \\ \mathrm{null}(D[:, 1 : k]) &= \mathrm{null}(A[:, 1 : k]) - 1 \\ \mathrm{null}(D[1 : k, :]^T) &= \mathrm{null}(A[1 : k, :]^T) - 1 \end{aligned}$$

So that Property 1 may not be satisfied by $D$. □

Please compare the result below with Corollary 1.

**Corollary 3.** *Matrix $A$ is of size $n$ and rank $r \leq n$. $A$ satisfies, for all $k = 1, \ldots, n$:*

$$\mathrm{null}(A[1:k, 1:k]) = \mathrm{null}(A[1:k, :]) \tag{4}$$

*if and only if $A$ has an LU factorization $A = LU$ where $U$ is unit upper triangular.*

*If $A$ satisfies Equation (4), then there exist a row permutation $P_r$ such that:*

$$B = P_r^{-1} A$$

*$B$ has an LU factorization: $B = L_B U_B$ where $U_B$ is unit upper triangular. Then:*

$$A = (P_r L_B) U_B = LU$$

*is an LU factorization of $A$ where $U = U_B$ is unit upper triangular.*

*We also have that column $i$ of $B$ (or column $i$ of $A$ since the columns are not permuted) is a linear combination of the previous columns if and only if row $i$ of $B$ is a linear combination of the previous rows.*

*Assume $e_{i_0} = P_r e_i$. If column $i$ of $B$ is a linear combination of the previous columns, then $i_0 \leq i$. We have $U[i, :] = e_i$, $L[:, i] = 0$, and $L[i_0, i_0 :] = 0$. If column $i$ of $B$ is linearly independent of the previous columns then $i_0 \geq i$ and $L[i_0, i+1 :] = 0$.*

*Proof.* Consider $B = A^T$. Apply Theorem 6. ☐

# A Ancillary results

We prove a few additional results that are helpful in the main proofs.

**Lemma 2.** *Assume that $x \in S = \mathrm{span}\{x_1, \ldots, x_r\}$. Denote by $X = [x_1, \ldots, x_r]$. Assume that $X[1:r, :]$ is non-singular. Then:*

$$x = X X[1:r, :]^{-1} x[1:r]$$

*Proof.* Since the submatrix $X[1:r, :]$ is non-singular, any vector $x \in S$ is uniquely determined by its first $r$ entries. Let $P$ be the oblique projection onto $S$ along the subspace of vectors with the first $r$ entries equal to zero. The projection matrix is given by:

$$P = X(Y^T X)^{-1} Y^T$$

where the range of $Y$ is the orthogonal complement of the null space of $P$. Specifically, we set:

$$Y = \begin{pmatrix} I_r \\ 0 \end{pmatrix}$$

Since $x \in S$, we have $x = Px$. Substituting the expressions for $P$ and $Y$, we obtain:

$$x = X(Y^T X)^{-1} Y^T x = X(X[1:r, :])^{-1} x[1:r]$$

☐

**Lemma 3.** *Consider a matrix $A$ of size $n \times k$, $k \leq n$. Assume that the rank of $A$ is $k-1$ and that column $k$ is linearly dependent on the previous columns. If $A$ has an LU factorization $A = LU$ with $u_{kk} = 1$, then column $k$ of $L$ is 0.*

*Proof.* Decompose $L$ and $U$ in blocks:

$$L = \begin{pmatrix} L_{11} & 0 \\ L_{21} & l_{kk} \end{pmatrix}, \quad U = \begin{pmatrix} U_{11} & u_{1k} \\ 0 & 1 \end{pmatrix}$$

where $L_{11}$ is of size $(k-1) \times (k-1)$, $U_{11}$ is of size $(k-1) \times (k-1)$, $L_{21}$ is of size $(n-k+1) \times (k-1)$, and $u_{1k}$ is of size $(k-1) \times 1$. Then:

$$l_{kk} = A[k:, k] - A[k:, 1:k-1] A[1:k-1, 1:k-1]^{-1} A[1:k-1, k]$$

By Lemma 2, since column $k$ of $A$ is linearly dependent on the previous columns, we have: $l_{kk} = 0$. ☐

**Corollary 4.** *Consider a matrix $A$ of size $k \times n$, $k \leq n$. Assume that the rank of $A$ is $k-1$ and that row $k$ is linearly dependent on the previous rows. If $A$ has an LU factorization $A = LU$ with $l_{kk} = 1$, then row $k$ of $U$ is 0.*

*Proof.* Consider $A^T = U^T L^T$. Apply Lemma 3. □

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